L D COLLEGE OF ENGINEERING

Computer Engineering Department

DESIGN AND DEVELOPMENT OF A FUZZY CONTROLLER FOR SMART TRAFFIC SIGNAL CONTROLLER

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Problem Identification

- PROBLEM :SMART TRAFFIC SIGNAL CONTROLLER
- Traditional traffic signals use fixed-time control.
- Traffic volume changes continuously during the day.
- Fixed green time can cause long waiting queues.
- It leads to traffic congestion, fuel consumption, and pollution.
- An intelligent system is required to adjust signal timing according to real-time traffic conditions.
- Proposed Solution: Use a Fuzzy Logic Controller to determine the required green signal time.

Objectives

- To develop an intelligent traffic signal control system using fuzzy logic.
- To adjust green signal time automatically according to traffic conditions.
- To reduce vehicle waiting time and traffic congestion.
- To improve traffic flow and intersection efficiency.
- To reduce unnecessary fuel consumption and air pollution.
- To provide a flexible alternative to fixed-time traffic signals.

Proposed Fuzzy Controller

Smart Traffic Signal Controller

The proposed system uses Fuzzy Logic to automatically adjust the green signal time based on traffic conditions.

Inputs:

- Traffic Density → 0–100%
- Waiting Time → 0–120 seconds

Output:

- Green Signal Time → 10–90 seconds
- Working Process
- Traffic Data → Fuzzification → Fuzzy Rule Base → Defuzzification → Green Signal Time

System Design

Inputs and Output

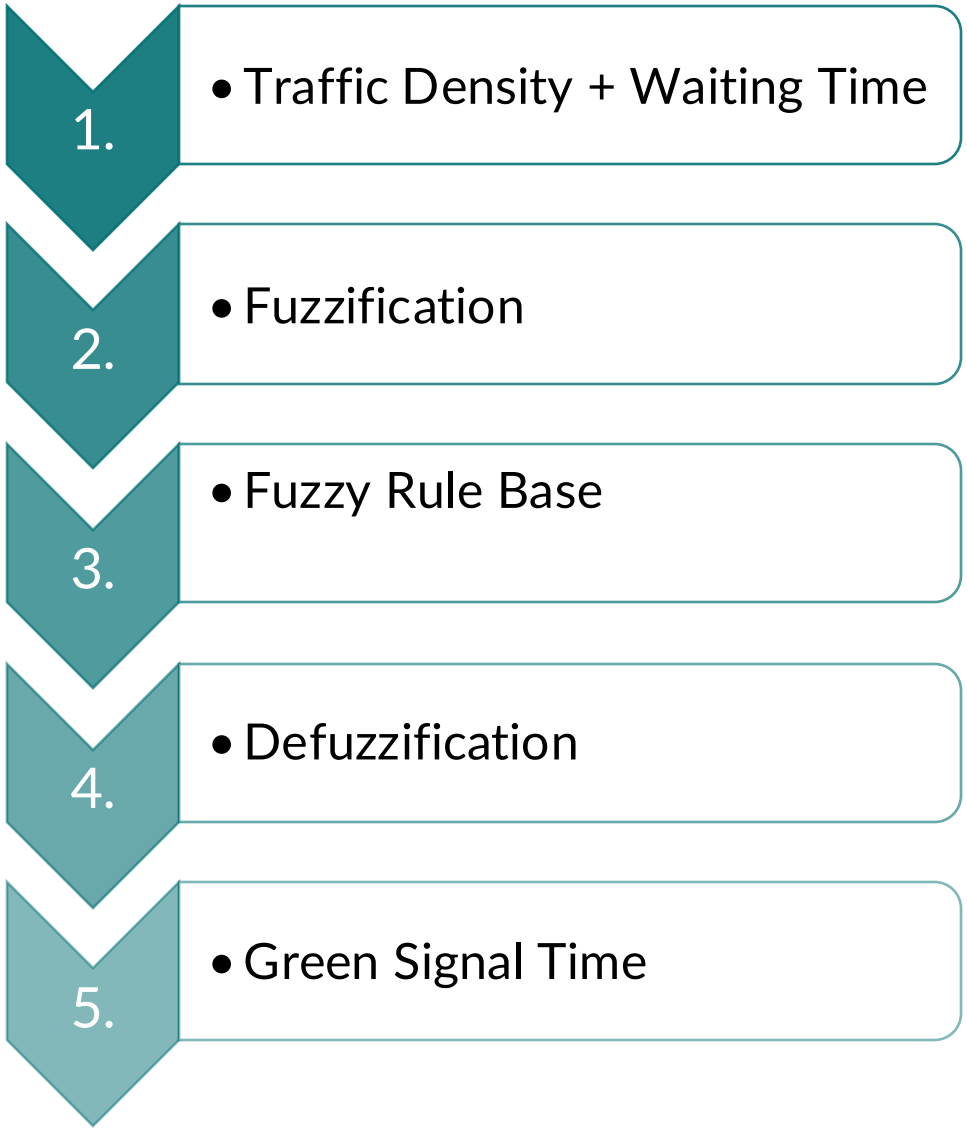
Inputs:

1. Traffic Density – 0 to 100%
2. Waiting Time – 0 to 120 seconds

Output:

1. Green Signal Time – 10 to 90 seconds

➢ Fuzzy Controller Process:



Input 1: Traffic Density

❖ Traffic Density

- Traffic density represents the amount of traffic present on a road at a particular time.
- Range: 0–100%

❖ Membership Function

- Triangular Membership Function
- Low → Medium → High
- A higher traffic density results in a requirement for longer green signal time.

Linguistic Variable	Range
Low (L)	0–50%
Medium (M)	25–75%
High (H)	50–100%

Input 2 : Waiting Time

❖Waiting Time

Waiting time represents the time for which vehicles wait at the traffic signal.

Range: 0–120 seconds

❖ Membership Function

Triangular Membership Function

Short → Medium → Long

Higher waiting time indicates a need for longer green signal duration.

Linguistic Variable	Range
Short (S)	0–60 sec
Medium (M)	30–90 sec
Long (L)	60–120 sec

Output: Green Signal Time

❖ Green Signal Time

The fuzzy controller determines the required green signal duration based on traffic density and waiting time.

Output Range: 10–90 seconds

❖ Membership Function

Triangular Membership Function

Very Short → Short → Medium → Long → Very Long

Linguistic Variable	Green Time
Very Short (VS)	10 sec
Short (S)	30 sec
Medium (M)	50 sec
Long (L)	70 sec
Very Long (VL)	90 sec

Fuzzy Rule Base

- Fuzzy Rules for Traffic Signal Control
- The controller uses IF–THEN rules to determine the green signal time.

Rule	Traffic Density	Waiting Time	Green Time
R1	Low	Short	Very Short
R2	Low	Medium	Short
R3	Low	Long	Medium
R4	Medium	Short	Short
R5	Medium	Medium	Medium
R6	Medium	Long	Long
R7	High	Short	Medium
R8	High	Medium	Long
R9	High	Long	Very Long

Example:IF Traffic Density is High AND Waiting Time is Long, THEN Green Time is Very Long.

Sample Data

Sample Input for Fuzzy Controller

For testing the proposed system, consider:

- Traffic Density = 65%
- Waiting Time = 75 seconds

Membership Values

Traffic Density (65%)

- Medium = 0.4
- High = 0.6

Waiting Time (75 sec)

- Medium = 0.5
- Long = 0.25

These membership values are used to identify the fired fuzzy rules and calculate their firing strengths.

Fired Rules & Firing Strength

Firing Strength Calculation

For the sample input:

Traffic Density = 65%

Waiting Time = 75 sec

Using AND = Minimum method:

Rule	Calculation	Firing Strength	Output
R5	min(0.4, 0.5)	0.40	Medium
R6	min(0.4, 0.25)	0.25	Long
R8	min(0.6, 0.5)	0.50	Long
R9	min(0.6, 0.25)	0.25	Very Long

Fired Rules: R5, R6, R8 and R9
Higher firing strength → Greater contribution to final green time.

Defuzzification & Crisp Output

- Weighted Average Method
- Green Time = $\Sigma(\alpha i \times Zi) / \Sigma \alpha i$

$= [(0.40 \times 50) + (0.25 \times 70) + (0.50 \times 70) + (0.25 \times 90)] / (0.40+0.25+0.50+0.25)$

$= 95 / 1.40$

$= 67.86$ seconds

Final Crisp Output:

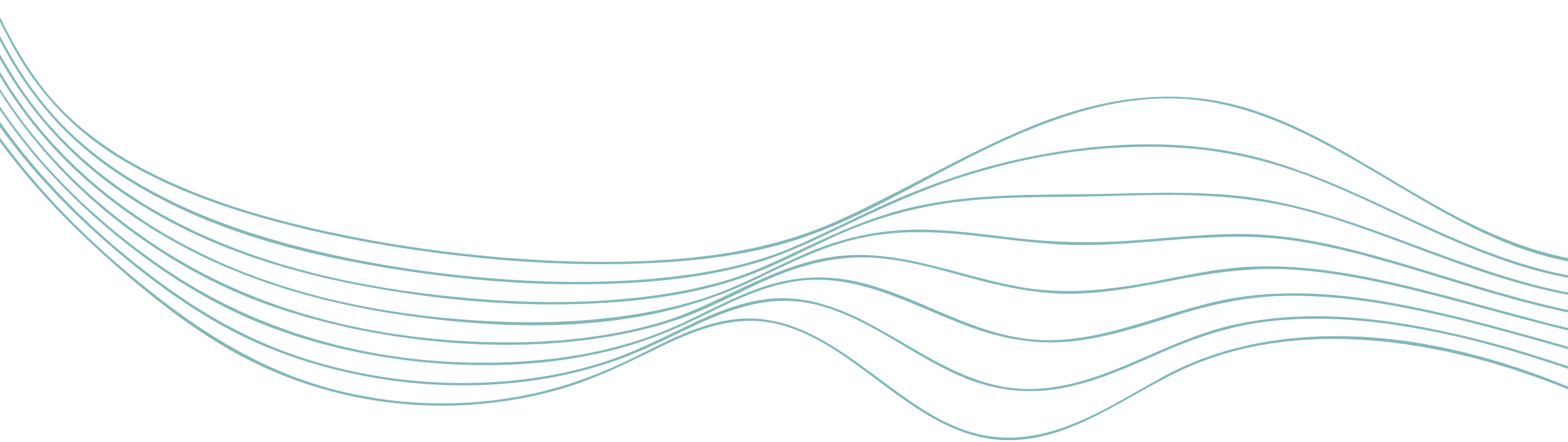
- Green Signal Time ≈ 68 seconds
- For 65% traffic density and 75 seconds waiting time, the fuzzy controller gives approximately 68 seconds of green signal time.

Conclusion

- The fuzzy controller provides an intelligent solution for traffic signal control.
- It uses traffic density and waiting time as input parameters.
- The green signal time is automatically adjusted according to traffic conditions.
- The system can reduce vehicle waiting time and traffic congestion.
- It can improve traffic flow at road intersections.
- It can help reduce fuel consumption and air pollution.
- Fuzzy logic is suitable for handling uncertain and changing traffic conditions.

Final Result

- For 65% traffic density and 75 seconds waiting time → Green Signal Time ≈ 68 seconds.



THANKS